



Full length article

Synthesis of Cu–Al hydrotalcite-SBA-15 composites and CO₂ capture using the sorbentChangchun Ji^{a,c}, Yudong Wang^{a,b,*}, Ning Zhao^{c,d,**}^a Shanxi Institute of Energy, 63 University Street, College Park, Jinzhong, 030600, PR China^b College of Textile Engineering, Taiyuan University of Technology, 79 Yingze West Street, Taiyuan 030024, PR China^c State Key Laboratory of Coal Conversion, Institute of Coal Chemistry, Chinese Academy of Sciences, Taiyuan 030001, PR China^d National Engineering Research Center for Coal-based Synthesis, Taiyuan, Shanxi, PR China, 030001

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ABSTRACT

Cu–Al hydrotalcite-SBA-15 composite materials are prepared by precipitation-deposition method. The CO₂ adsorption and regeneration process are carried out in a fixed bed reactor. This composite has thermal stability derived from hydrotalcite and moderate regeneration temperature from the porous material SBA-15. The adsorption capacities of the composite are obviously improved compared to the pure SBA-15 or hydrotalcite. When the amount of hydrotalcite is increased to 40 wt%, it has the magnitude of adsorption capacity. The inactivation model can well simulate the CO₂ adsorption process of the composite.

1. Introduction

Global warming caused by carbon dioxide (CO₂) emissions has become a dramatic impact on the Earth's environment [1]. The carbon dioxide emitted into the air is mainly from the burning of fossil fuels [2]. Many research groups in different fields have been working to reduce carbon dioxide emissions, and developing effective methods for capturing carbon dioxide from flue gas. There are several techniques for removing carbon dioxide from different industrial streams: adsorption, absorption, membrane, cryogenic separation [3–8]. Compared to solvent absorption and membrane separation techniques, solid adsorption has several potential advantages, such as high capacity, good selectivity and non-corrosive equipment. In recent work, solid adsorption technology has been considered to capture carbon dioxide [9].

According to the working temperature, solid adsorbent can be divided into low temperature adsorbents, medium temperature adsorbents and high temperature adsorbents. Most of the low temperature adsorbents such as carbon-based materials, micro-mesoporous silica, molecular sieves and organic frameworks (MOF) are porous materials [10–14]. These materials have potential applications for CO₂ capture due to many advantages such as low density, high specific surface areas and high porosity. Microporous materials include activated carbon, carbon nanotube, x-type molecular sieves, A-type molecular sieve, zeolite, etc. [15–25]. Ordered mesoporous materials MCM-41 and SBA-15, have attracted attention due to their outstanding high specific

surface area, adjustable pore size and ordered pore structure [26–28]. However, since the mesoporous material has thin-walled holes and amorphous holes, it is liable to collapse under high temperature or humidity conditions. High temperature adsorbents are a class of adsorbents operating at temperatures above 400 °C, such as lithium zirconates, orthosilicates, CaO and hydrotalcite [29–32]. Compared with low temperature adsorbents and medium temperature adsorbents, high temperature adsorbents have the advantage of high thermal stability. However, the disadvantage is high temperature regeneration which makes industrial applications difficult. It has been found that among these high temperature adsorbents, the hydrotalcite-based adsorbent has a relatively low regeneration temperature and good thermal stability [33,34]. However, its CO₂ adsorption performance is not excellent due to its small specific surface area. Therefore, many researchers are working to increase the adsorption amount of hydrotalcite by different methods. Gang et al. introduced Fe(CN)₆^{4–} in hydrotalcite synthesis, forming Mg²⁺-Al³⁺-Fe(CN)₆^{4–} [35]. By introducing Fe(CN)₆^{4–}, the ratio of Al³⁺/(Mg²⁺ + Al³⁺) was improved, which increased the adsorption amount of CO₂. Wang et al. also confirmed this conclusion by studying the material Mg-M-CO₃ (M = Al, Fe, Ga, Mn). These methods focus on ion exchange of interlayer ions of hydrotalcite. There are few studies on the mesoporous LDHs system [36]. Chang et al. obtained CaAILDHs with a specific surface area of 270 m²g^{–1} and a nano-sized structure by high-temperature calcination, which increased the adsorption amount of CO₂ [37]. However, this material did not

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work at low temperatures. In 2013, in order to achieve high CO₂ adsorption of hydrotalcite under low temperature condition, Pramod proposed the use of mesoporous materials as carriers to support hydrotalcite [38].

In the study, using the high specific surface area mesoporous material as a carrier, the mesoporous material and the layered adsorbent material can be combined to prepare a new material Cu–Al hydrotalcite-SBA-15. The composite material has an high CO₂ adsorption capacity at moderate regeneration temperatures.

2. Experimental

2.1. Material preparation

2.1.1. Preparation of SBA-15

The mesoporous material SBA-15 was synthesized based on the sol-gel method reported by Zhao et al. [39]. First, a certain amount of the templating agent EO₂₀-PO₇₀-EO₂₀ (P123, Sigma-Aldrich) was dissolved in 95 g of water with stirring and dissolved in a 35 °C solution of water bath. When, the templating agent was completely dissolved, hydrochloric acid was added. Next, the tetraethyl orthosilicate (TEOS, Aladdin) was added dropwise with a separatory funnel into the sample which was stirred uniformly. Then, the solution was further stirred for 4 h, and the formed sol was transferred to a polytetrafluoroethylene reactor and aged at 120 °C for 48 h in teflon lined sealed container. Finally, in the air atmosphere, a powdery mesoporous SBA-15 material was obtained by calcination at 550 °C for 6 h at a heating rate of 2 °C·min⁻¹, which was used as a carrier in later experiments.

2.1.2. Preparation of CuAl hydrotalcite

The CuAl hydrotalcite was synthesized by coprecipitation. The crystals of Cu(NO₃)₂·3H₂O, and Al(NO₃)₃·9H₂O were weighed in a certain molar ratio of 1:2, 1:3, 1:4, dissolved in deionized water to prepare a mixed salt solution, and charged into a separatory funnel. Similarly, Na₂CO₃ and NaOH were dissolved in deionized water at a certain molar ratio to prepare a mixed alkali solution, which was poured into a separatory funnel. The mixed base and the mixed salt were then slowly co-dried into the beaker and stirred vigorously, and the pH of the liquid in the beaker was always maintained in the range of 9–10 during the co-drip. Finally, the mixed solution was stirred for a further 40 min, then the reaction was aged in an 80 °C incubator for 18 h, cooled, washed with deionized water until the pH of the supernatant was 7. The sample was dried for 8 h to obtain CuAl Hydrotalcite, named CuxAl-HTLcs, x was the molar ratio of Cu/Al.

2.1.3. Preparation of the composites

A certain amount of the carrier SBA-15 synthesized in 2.1.1 was added to a beaker of a certain amount of water, and uniformly stirred in a 40 °C water bath for 3 h. Then, CuAl-based hydrotalcite was synthesized on the SBA-15 carrier in the above beaker according to the method of synthesizing CuAl-based hydrotalcite in 2.1.2. CuAl-based hydrotalcite material which was used for synthetic hydrotalcite would be added to the solution. They were mixed under vigorous stirring at 40 °C, and the addition speed of the two mixed solutions was adjusted to obtain a constant pH of 9. The gel was aged at 80 °C for 18 h. The product was filtered and washed thoroughly with water so that a gel of pH 7 was obtained. The sample was named HTLcs-SBA-15-x, where x was the percentage of HTLcs.

2.2. Measurements of CO₂ adsorption capacity

The evaluation of the adsorption capacity of the solid adsorbent for CO₂ was carried out on a fixed bed reactor. Under normal pressure and a certain desorption temperature, the adsorbent was purged and activated under an argon atmosphere for 2 h, which the volume flow rate of the argon gas was always controlled to 50 ml·min⁻¹. When the sample

was purged, the temperature of the fixed bed was lowered to the desired adsorption temperature, and the inlet gas was switched to the simulated flue gas. The Vaisala gas analyzer was used to simultaneously detect the CO₂ concentration in the outlet gas, and the recording frequency was 10s/time until the CO₂ content in the exhaust gas and the CO₂ content in the mixed gas at the inlet were consistent. The above evaluation of the adsorption performance of the solid adsorbent CO₂ was completed.

2.3. Characterization

The XRD (X-ray diffraction) was measured on an X'pert Pro X-ray diffractometer from PANalytical (Netherlands) with Cu Kα radiation ($\lambda = 1.5406 \text{ \AA}$).

The specific surface area, pore volume and average pore size were determined on a Micromeritics ASAP 2020 physical adsorber. All adsorbent samples were degassed under vacuum at 120 °C prior to testing.

The structure and morphology of the sample were analyzed on a HRTEM model of JEM-2100 by JEOL.

Elemental chemical analysis of Cu, Al was carried out by atomic absorption on a vario Macro cube apparatus (Elementar).

3. Results and discussion

3.1. Effect of different Cu/Al ratios on hydrotalcite

The structural parameters of the three samples with different Cu/Al ratios, surface area (BET) and pore volume (Cu/Al = 2:1, 3:1, 4:1) are listed in Table 1. It can be seen from the table that the actual value of the Cu/Al molar ratio is very close to the theoretical value, indicating that the hydrotalcite having the ideal Cu/Al molar ratio is synthesized, and is replaced by a theoretical value in the following description. Therefore, in the following description, the theoretical molar ratio of Cu/Al directly replaces the experimental value. Moreover, as the Cu/Al molar ratio increases, the specific surface area and pore volume of the material are significantly reduced.

Fig. 1 shows the XRD spectra of the CuxAl-HTLcs with different Cu/Al ratios. It can be seen from the figure that the hydrotalcite-like peaks of the three samples have obvious diffraction peaks of hydrotalcite at $2\theta \approx 11.4^\circ, 22.8^\circ, 34.8^\circ$, indicating that all three samples have hydrotalcite structure [40–43]. However, from the viewpoint of the intensity of the characteristic peak, as the molar ratio of Cu/Al increases, the intensity of the hydrotalcite-like characteristic peak at $2\theta \approx 11.4^\circ$ decreases, which means that the crystallinity of the hydrotalcite decreases. Further, a material having a Cu/Al molar ratio of 2 has a relatively large specific surface area and pore volume. Therefore, the Cu/Al molar ratio is fixed to 2 in the synthesis of the composite material in the future, and the hydrotalcite having the Cu/Al molar ratio is fixed and simplified as HTLcs.

3.2. Effect of loading of HTLcs on the mesoporous material SBA-15

The small and wide-angle spectra of the composite are shown in Fig. 2. It can be seen from the small angle spectrum that when a certain amount of hydrotalcite HTLcs is deposited on the mesoporous SBA-15, the composite still has a certain mesoporous structure. However, the

Table 1
Structure parameters of CuxAl-HTLcs with different Cu/Al ratios.

Cu/Al theoretical value Molar ratio	Cu/Al experimental value Molar ratio	S _{BET} (m ² ·g ⁻¹)	Pore Volume (cm ³ ·g ⁻¹)
2	2	59.3	0.146
3	3.1	45.9	0.108
4	4.1	36.2	0.056

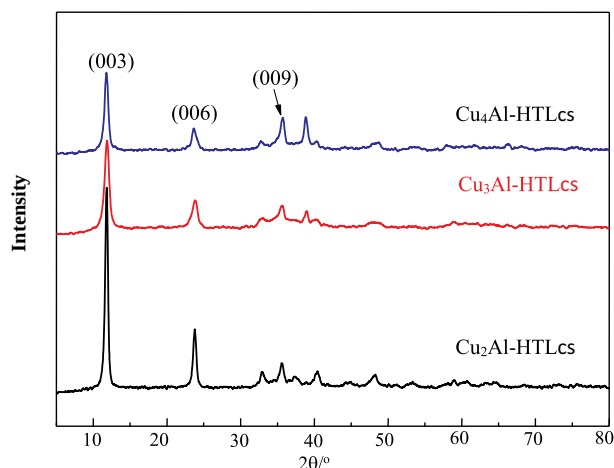


Fig. 1. XRD patterns of $\text{Cu}_x\text{Al-HTLcs}$ with different Cu/Al ratios.

loading of HTLcs reached 50 wt%, the mesoporous characteristic peak only has a characteristic peak at $2\theta \approx 1.0^\circ$, and other characteristic peaks almost completely disappear. It can be an amorphous form of SBA-15 where the signal does not reach the XRD detection limit [38,43,44]. In addition, it can be seen from the wide-angle spectrum that when the hydrotalcite loading reaches 40% by weight, the characteristic peak of the composite material is consistent with the single hydrotalcite material, and at this time, the structure of the mesoporous support is not destroyed. When the loading is low, the diffraction peak of the hydrotalcite does not appear because the structure of the hydrotalcite is not formed, or the amount formed is too low. Therefore, at a loading of 40% by weight, the composite material has both a stable mesostructured support and a complete hydrotalcite structure [38,45].

Fig. 3 shows the N_2 adsorption-desorption test of the composite HTLcs-SBA-15- x . The structural parameters of the material calculated by the isotherm are listed in Table 2. It can be seen from the figure that the mesoporous material isotherm belongs to type IV and the H1 hysteresis loop, indicating that the carrier has uniform mesopores [39]. The simple hydrotalcite hysteresis ring is a typical H3 type, and the pore diameter is mainly a slit hole formed by tabular grains. When the loading of HTLcs is small, the mesoporous carrier can maintain the H1 type, indicating that the material still has a uniform mesopores. However, when the loading exceeds 40% by weight, the H1 type hysteresis loop is converted to the H3 type [42,43]. Moreover, it can also be seen

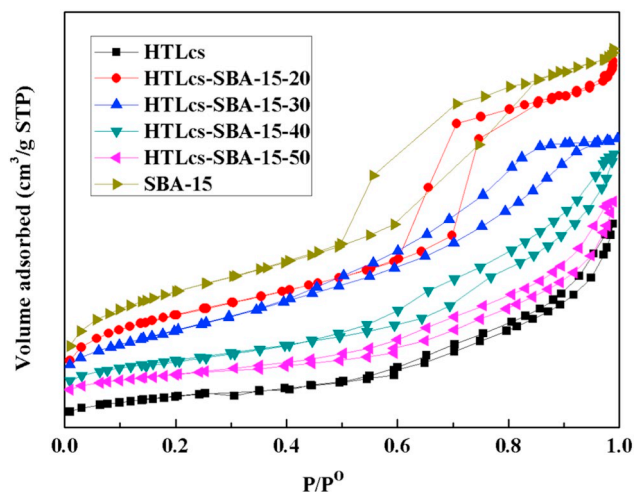


Fig. 3. N_2 adsorption-desorption isotherms of the composites HTLcs-SBA-15- x .

Table 2

The structural properties of the composites HTLcs-SBA-15- x .

Sample	$S_{\text{BET}}(\text{m}^2\text{g}^{-1})$	$V_{\text{BET}}(\text{cm}^3\text{g}^{-1})$	$D_m(\text{nm})$
SBA-15	647	1.036	9.4
HTLcs-SBA-15-20	476	0.872	8.9
HTLcs-SBA-15-30	440	0.657	10.3
HTLcs-SBA-15-40	421	0.284	11.0
HTLcs-SBA-15-50	399	0.263	8.4
HTLcs	54	0.146	—

from Table 2 that as the loading amount increases, the specific surface area and pore volume of the material decrease. When the loading amount reaches 50 wt%, the specific surface area of the material decreases from $647 \text{ m}^2\text{g}^{-1}$ to $399 \text{ m}^2\text{g}^{-1}$, and the pore volume decreases from $1.036 \text{ cm}^3\text{g}^{-1}$ to $0.263 \text{ cm}^3\text{g}^{-1}$ [46–48]. Therefore, it can be concluded that hydrotalcite can enter most of the pores when the loading is small. However, when the load is excessive, the hydrotalcite will cover the surface of the mesopores [38,46].

In order to investigate the effect of hydrotalcite loading on the morphology of mesoporous materials, TEM analysis is performed on the materials, as shown in Fig. 4. It can be seen from the figure that SBA-15 is a two-dimensional hexagonal channel structure, and HTLcs presents a

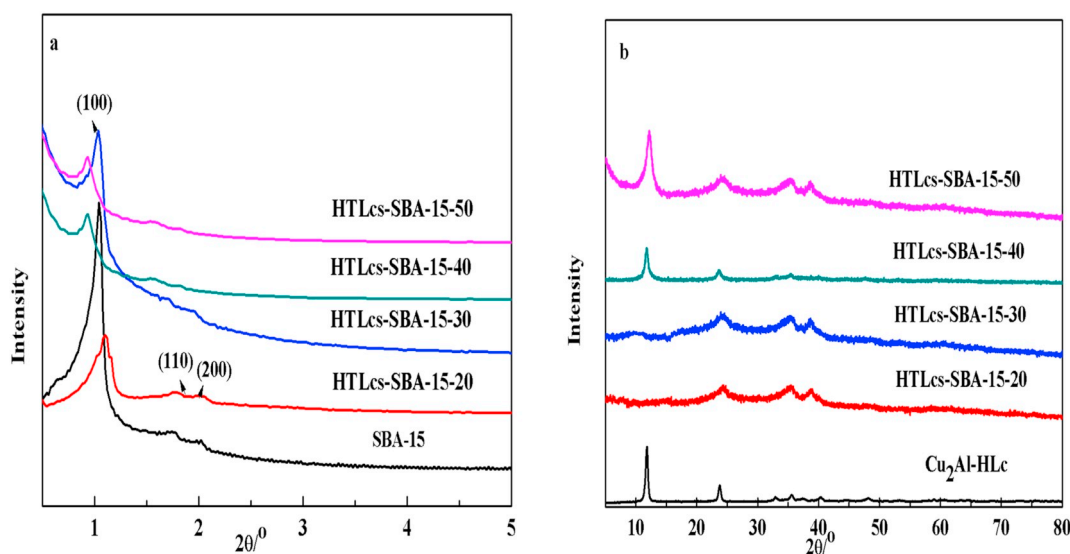


Fig. 2. (a) Low angle XRD patterns of the composites HTLcs-SBA-15- x ; (b) Wide angle XRD patterns of the composites HTLcs-SBA-15- x .

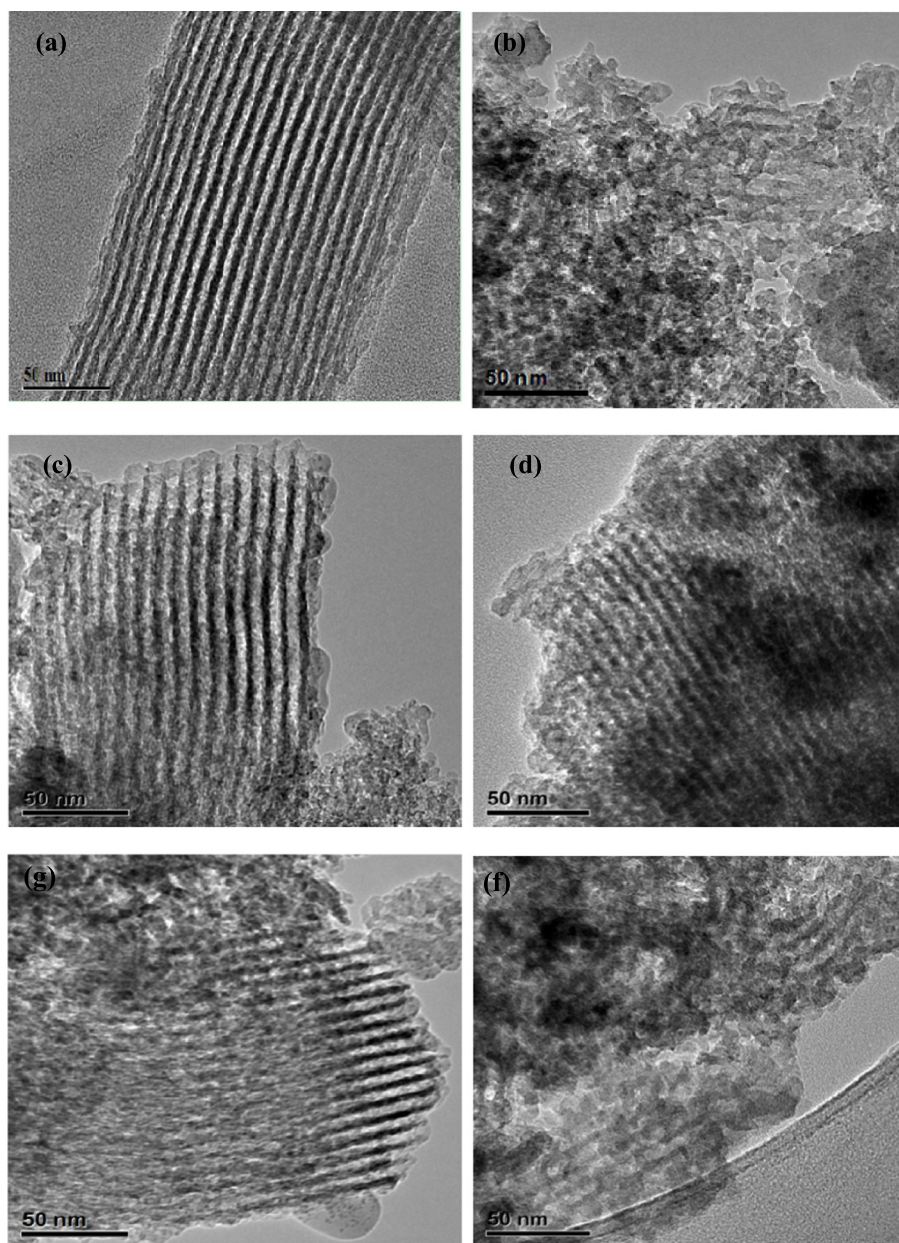


Fig. 4. Transmission electron microscope image of the composites. (a) SBA-15; (b) HTLcs; (c) HTLcs-SBA-15-20; (d) HTLcs-SBA-15-30; (e) HTLcs-SBA-15-40; (f) HTLcs-SBA-15-50.

slice structure [39,46]. The composites are investigated for loadings of 20 wt%, 30 wt%, 40 wt% and 50 wt%. When the hydrotalcite loading is low, it can be well dispersed into the mesopores, and the two-dimensional hexagonal structure of the mesopores can be well maintained. As can be seen from Fig. 4(e), when the loading amount reaches 40% by weight, the composite still maintains a regular mesoporous structure. However, when the loading is too high, most of the hydrotalcite covers the surface of the mesopores, as shown in Fig. 4(f) [38,46–50].

3.2.1. Effect of HTLcs loading on adsorption properties of composites

Fig. 5 shows the effect of different HTLcs loading on the adsorption performance of composite HTLcs-SBA-15-x. It can be clearly seen from the figure that the adsorption breakthrough time of CO₂ shows a trend of increasing first and then decreasing as the loading of HTLcs increases. The adsorption amount of CO₂ calculated from the breakthrough curve is shown in Table 3. It can be seen intuitively that when the loading of HTLcs is 40 wt%, the adsorption capacity of the carbon

dioxide of the composite increases to a maximum of 1.3 mmol·g⁻¹, because HTLcs are most dispersed in the mesoporous material. The increase in adsorption amount is achieved by the synergy of HTLcs and SBA-15. However, when the content of HTLcs continues to increase, excess HTLcs overflow from the pores and cover the outside of the mesoporous material, which increases the difficulty of diffusion of CO₂ into the pores, and thus the amount of adsorption decreases.

3.2.2. The deactivation model predictions

The adsorption process of CO₂ by HTLcs-SBA-15-x composites is also a non-catalytic gas-solid reaction. Therefore, we also simulate the adsorption breakthrough curve of the material by the deactivation model. At the same time, two parameters k_o and k_d are also used to simulate the data, and the applicability of the inactivation model is proved by the correlation coefficient R^2 .

The penetration curve obtained by the inactivation model is shown in Fig. 6. The k_o , k_d and R^2 obtained by simulating the experimental data

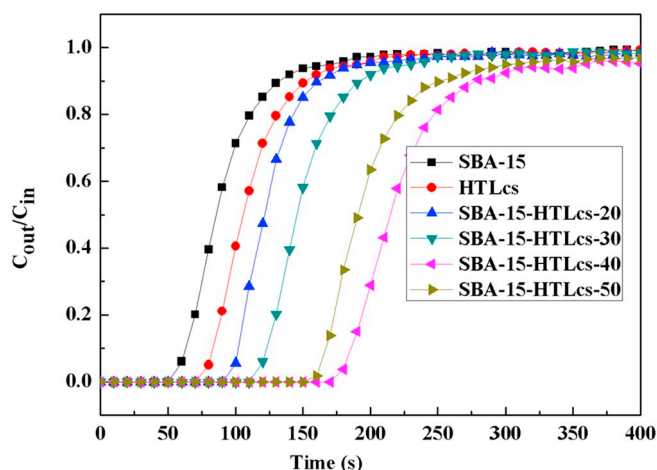


Fig. 5. Breakthrough curves of the composites HTLcs-SBA-15-x.

are listed in Table 4. It can be seen from the figure that the penetration curve is highly compatible with the experimental data, and the correlation coefficient R^2 is above 0.99, which proves that the inactivation model can well simulate the CO_2 adsorption process of the composite. It can also be seen from the table that the k_o and k_d of the mesoporous SBA-15 and hydrotalcite-like HTLcs are $165.30 \text{ cm}^3 \cdot \text{min}^{-1} \cdot \text{g}^{-1}$, $2.64 \text{ cm}^3 \cdot \text{min}^{-1} \cdot \text{g}^{-1}$ and $128.16 \text{ cm}^3 \cdot \text{min}^{-1} \cdot \text{g}^{-1}$, $2.82 \text{ cm}^3 \cdot \text{min}^{-1} \cdot \text{g}^{-1}$ respectively. When HTLcs is introduced into the mesopores, k_o increases to $309.20 \text{ cm}^3 \cdot \text{min}^{-1} \cdot \text{g}^{-1}$, and the increase is very significant; However, the increase in k_d was not much ($3.22 \text{ cm}^3 \cdot \text{min}^{-1} \cdot \text{g}^{-1}$). When the loading is 40% by weight, the composite has the largest k_o . This shows that the composite material achieves the maximum synergy between the two materials at this loading, which is consistent with the maximum CO_2 adsorption capacity of the composite with a loading of 40% by weight. However, when the loading amount is too high, although the adsorption rate is still increased, the desorption rate is greatly increased, so the adsorption amount of CO_2 is lowered.

3.2.3. Regenerability of HTLcs-SBA-15-40 sorbent

The adsorbent must satisfy one condition is that there is still a stable adsorption capacity after multiple adsorption desorption cycles. Therefore, the recoverability of the HTLcs-SBA-15-40 adsorbent is measured experimentally (Fig. 7). Even after 14 cycles, the sample has a stable adsorption capacity, indicating that the material has a certain cycle stability.

3.2.4. Effect of desorption temperature on adsorption properties of composites

Because the HTLcs is high-temperature CO_2 adsorbents, the desorption requires a high temperature, and the general desorption temperature is above 400°C . Therefore, it is necessary to investigate the influence of the desorption temperature of the composite on the adsorption performance of the material during the adsorption process. The CO_2 adsorption amounts of the composite HTLcs-SBA-15-40 at different desorption temperatures ($100\text{--}400^\circ\text{C}$) are listed in Table 5. It

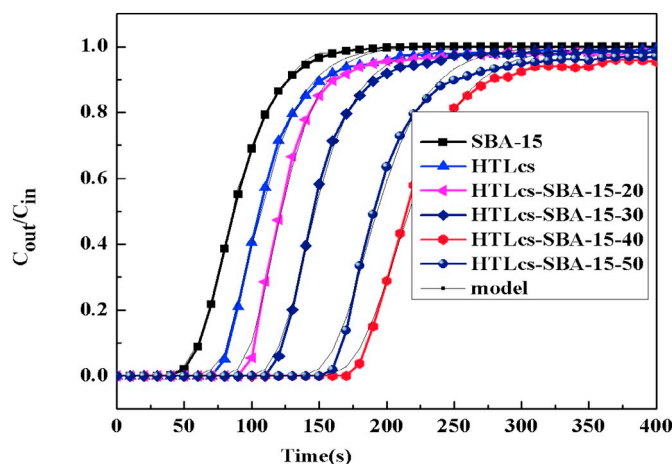


Fig. 6. Breakthrough curves of the composites and the lines correspond to de-activation model predictions.

Table 4

Rate Parameters of the composites from the breakthrough data.

Sample	k_o $\text{cm}^3 \cdot \text{min}^{-1} \cdot \text{g}^{-1}$	k_d $\text{cm}^3 \cdot \text{min}^{-1} \cdot \text{g}^{-1}$	R^2
SBA-15	165.30	2.64	0.9981
HTLcs-SBA-15-20	309.20	3.22	0.9988
HTLcs-SBA-15-30	326.11	2.81	0.9992
HTLcs-SBA-15-40	361.20	2.10	0.9984
HTLcs-SBA-15-50	364.06	2.39	0.9983
HTLcs	128.16	2.82	0.9986

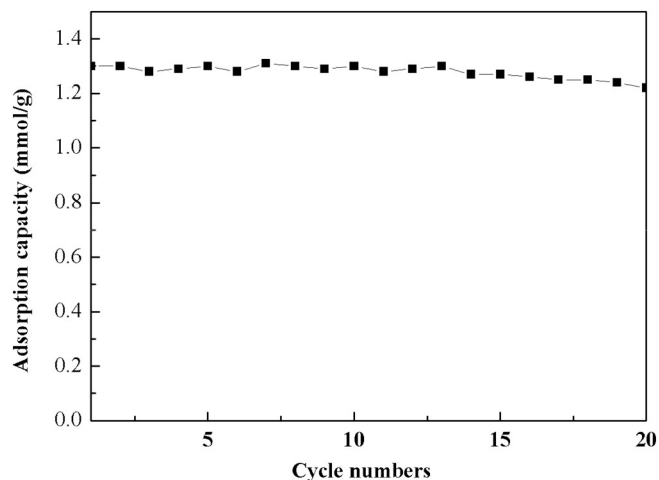


Fig. 7. Recyclability of HTLcs-SBA-15-40 sorbent.

is evident from the table that the material is completely desorbed at 150°C . Therefore, the synergy between HTLcs and SBA-15 allows the material to have a certain amount of adsorption at low temperatures,

Table 3

CO_2 adsorption capacity of the composites HTLcs-SBA-15-x.

Sample	Adsorption Temperature ($^\circ\text{C}$)	CO_2 (vol%)	N_2 (vol%)	Adsorption Capacity ($\text{mmol} \cdot \text{g}^{-1}$)
SBA-15	70	10	90	0.4
HTLcs-SBA-15-20	70	10	90	0.8
HTLcs-SBA-15-30	70	10	90	1.0
HTLcs-SBA-15-40	70	10	90	1.3
HTLcs-SBA-15-50	70	10	90	1.2
HTLcs	70	10	90	0.6

Table 5

CO₂ adsorption capacity of the composites HTLCs-SBA-15-40 under different desorption temperature.

Adsorption temperature (°C)	Desorption temperature (°C)	CO ₂ (vol%)	N ₂ (vol %)	Adsorption capacity (mmol·g ⁻¹)
70	100	10	90	0.8
70	150	10	90	1.3
70	200	10	90	1.3
70	250	10	90	1.3
70	300	10	90	1.3
70	350	10	90	1.3
70	400	10	90	1.3

and the desorption temperature is lowered.

4. Conclusions

The composite material HTLCs-SBA-15-x was synthesized by co-precipitation method using mesoporous material as SBA-15 as carrier. The effects of hydrotalcite-like HTLCs loading on mesoporous structure, morphology, pore size and CO₂ adsorption performance were investigated. It was proved by a series of characterization methods that HTLCs entered the mesopores. When the loading amount was 40 wt%, the composite material not only had a complete mesoporous structure but also had layered HTLCs in the pores. The synergistic effect of mesoporous SBA-15 and HTLCs increased the low temperature adsorption of the composite. The simulation of the experimental data by the in-activation model demonstrated that the introduction of HTLCs into the mesoporous material increased the adsorption rate of CO₂.

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Conflicts of interest

We declare that we have no conflict of interest.

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